COURSE PREREQUISITES USING TREE DATA STRUCTURE

Technical Presentation

CSA0313 – DATA STRUCTURES

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Problem Statement

- A university offers courses where students must complete prerequisite courses before enrolling in advanced subjects.
- **Example:**
- Programming Fundamentals → Data Structures → Algorithms → Artificial Intelligence
- **Objectives**
- Store prerequisite relationships efficiently.
- Display course hierarchy clearly.
- Verify student eligibility for advanced courses.
- Support easy curriculum updates.

Recommended Representation Structure

- **Tree Data Structure**
- A Tree is a hierarchical, non-linear data structure that represents parent-child relationships.
- **Components**
- **Root Node:** Foundation course
- **Parent Node:** Prerequisite course
- **Child Node:** Dependent course
- **Leaf Node:** Final course with no further prerequisites

Example Hierarchy

```
Programming Fundamentals
          |
   Data Structures
     /         \
Algorithms    Database Systems
     |
Artificial Intelligence
```

Each node represents a course, and each edge represents a prerequisite relationship.

Advantages

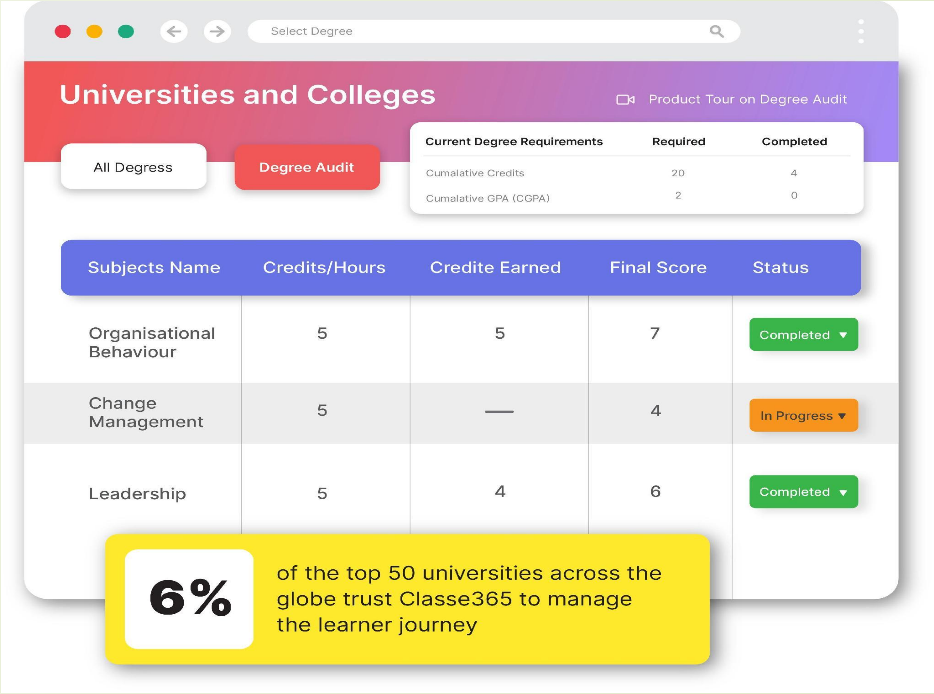
- Naturally represents hierarchical relationships.
- Easy to understand and visualize.
- Supports efficient traversal using **DFS** and **BFS**.
- Simplifies prerequisite verification.
- Easy to insert, update, or remove courses.
- Reduces redundancy by storing each course only once.

Benefits

- Organized course structure
- Faster academic planning
- Scalable for future curriculum expansion

Applications

- Course Registration System
- Degree Audit
- Curriculum Planning
- Student Eligibility Checking
- Learning Management Systems (LMS)



Conclusion

- A Tree Data Structure is the most suitable representation for hierarchical course prerequisites.
- It provides a clear and organized structure for storing course dependencies.
- Tree traversal enables efficient prerequisite checking and academic planning.
- It is easy to maintain and expand as new courses are added.
- If a course has multiple prerequisites, a Directed Acyclic Graph (DAG) is a better alternative.

Code of Conduct:

I S. Bhavya (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness